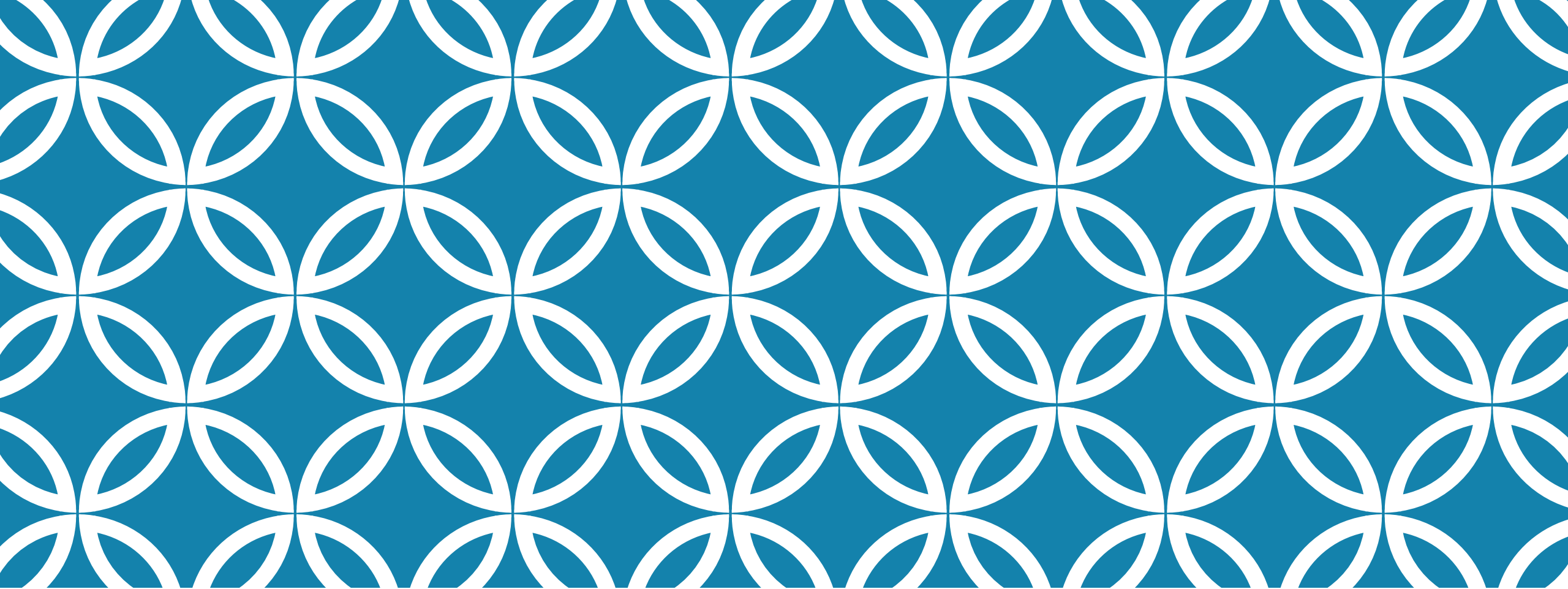




COMPUTER PROGRAMMING WEEK 7

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- Publish material by Robert Lafore.

INHERITANCE

A child inherits characteristics of its parents

Besides inherited characteristics, a child may have its own unique characteristics

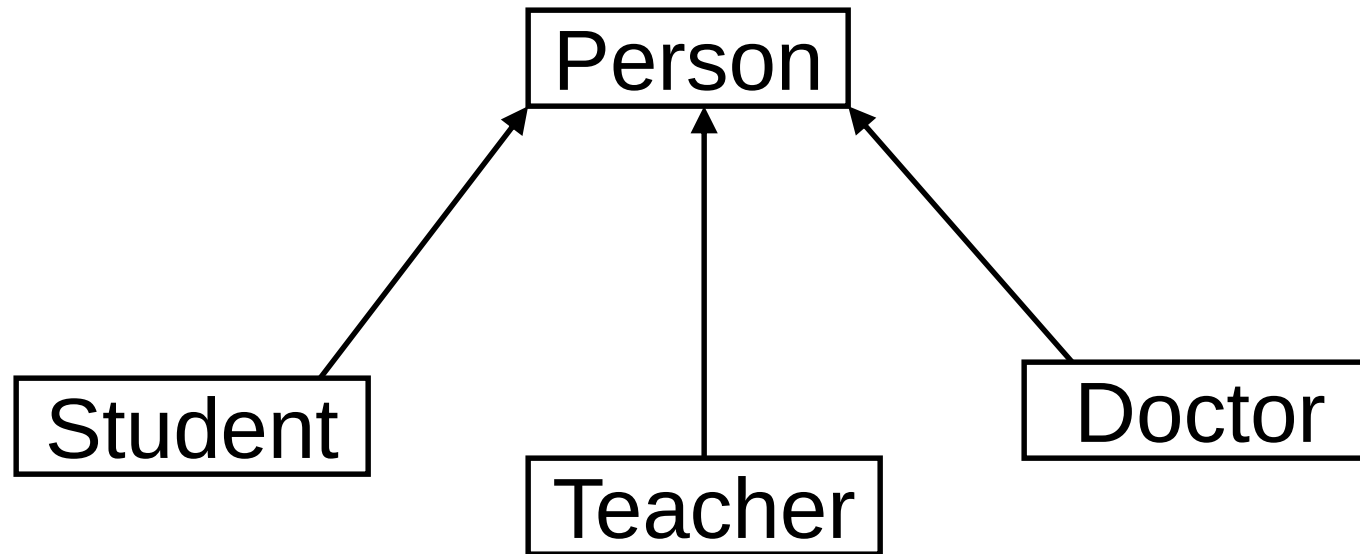
INHERITANCE IN CLASSES

If a class B inherits from class A then it contains all the characteristics (information structure and behaviour) of class A

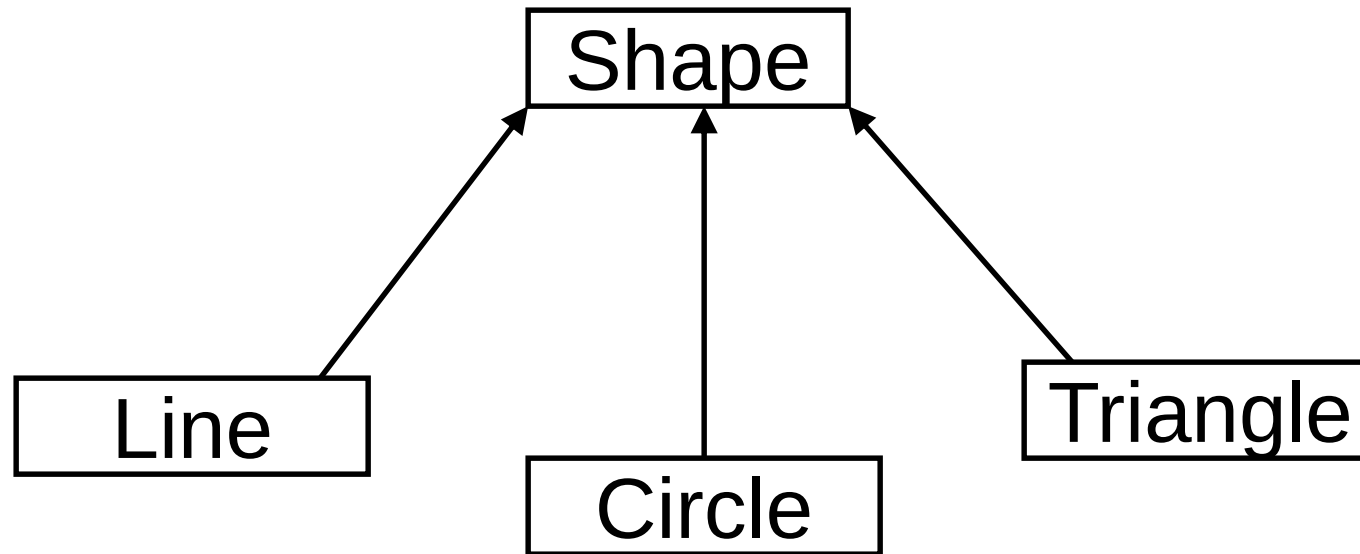
The parent class is called *base class* and the child class is called *derived class*

Besides inherited characteristics, derived class may have its own unique characteristics

EXAMPLE — INHERITANCE



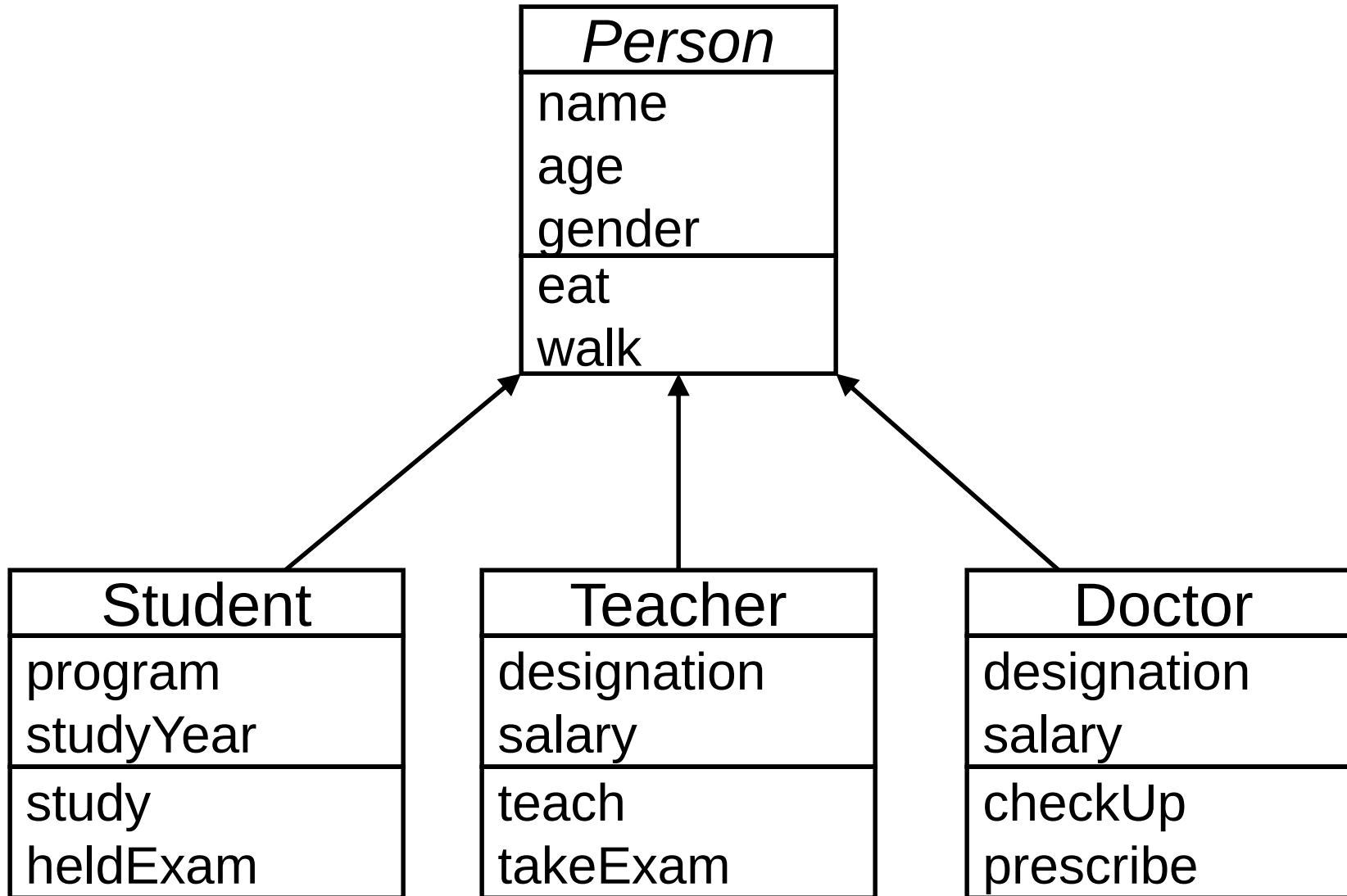
EXAMPLE — INHERITANCE



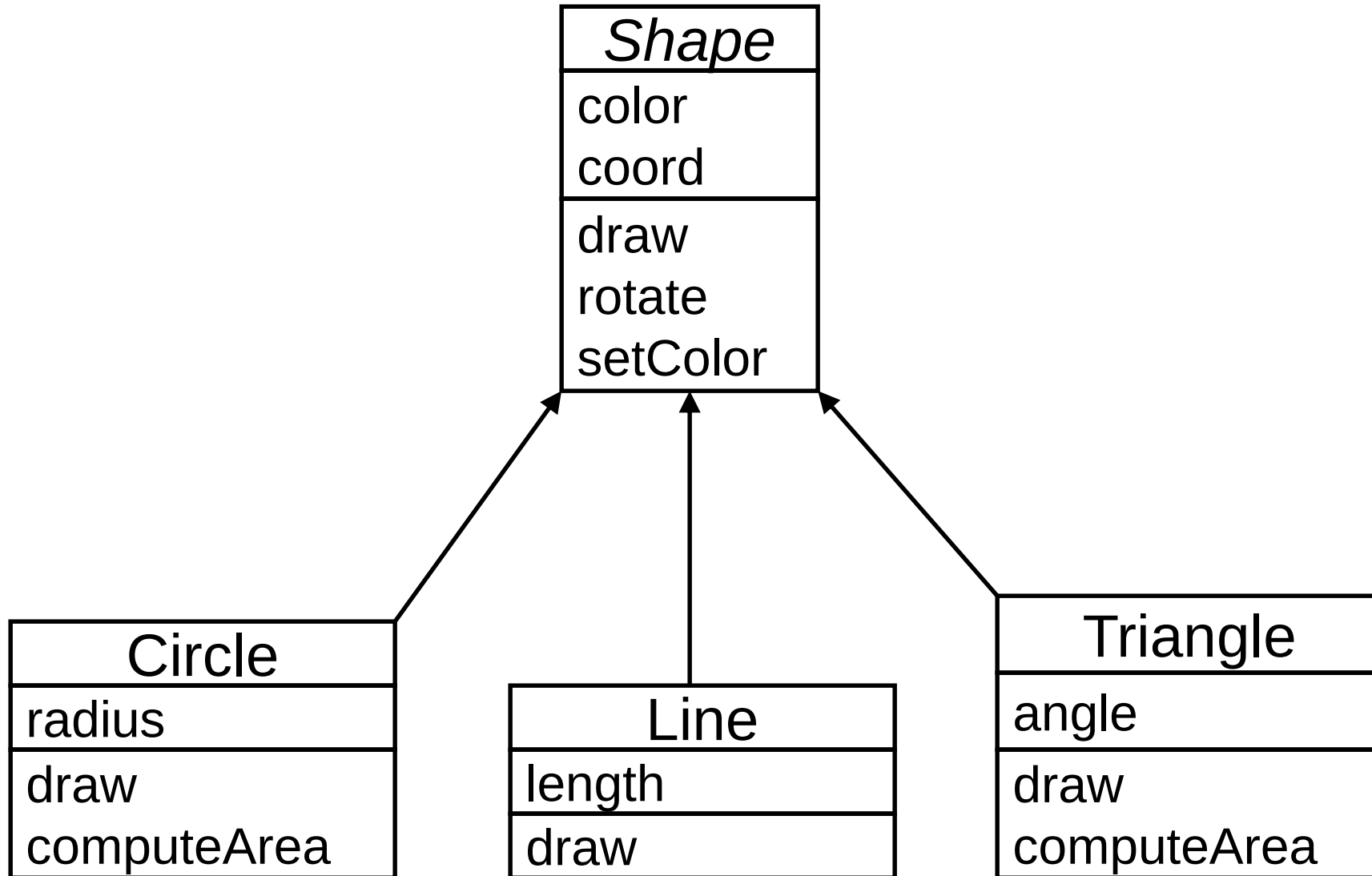
INHERITANCE — “IS A” OR “IS A KIND OF” RELATIONSHIP

Each derived class is a special kind of its base class

EXAMPLE — “IS A” RELATIONSHIP



EXAMPLE — “IS A” RELATIONSHIP





INHERITANCE — ADVANTAGES

Reuse

Less redundancy

Increased maintainability

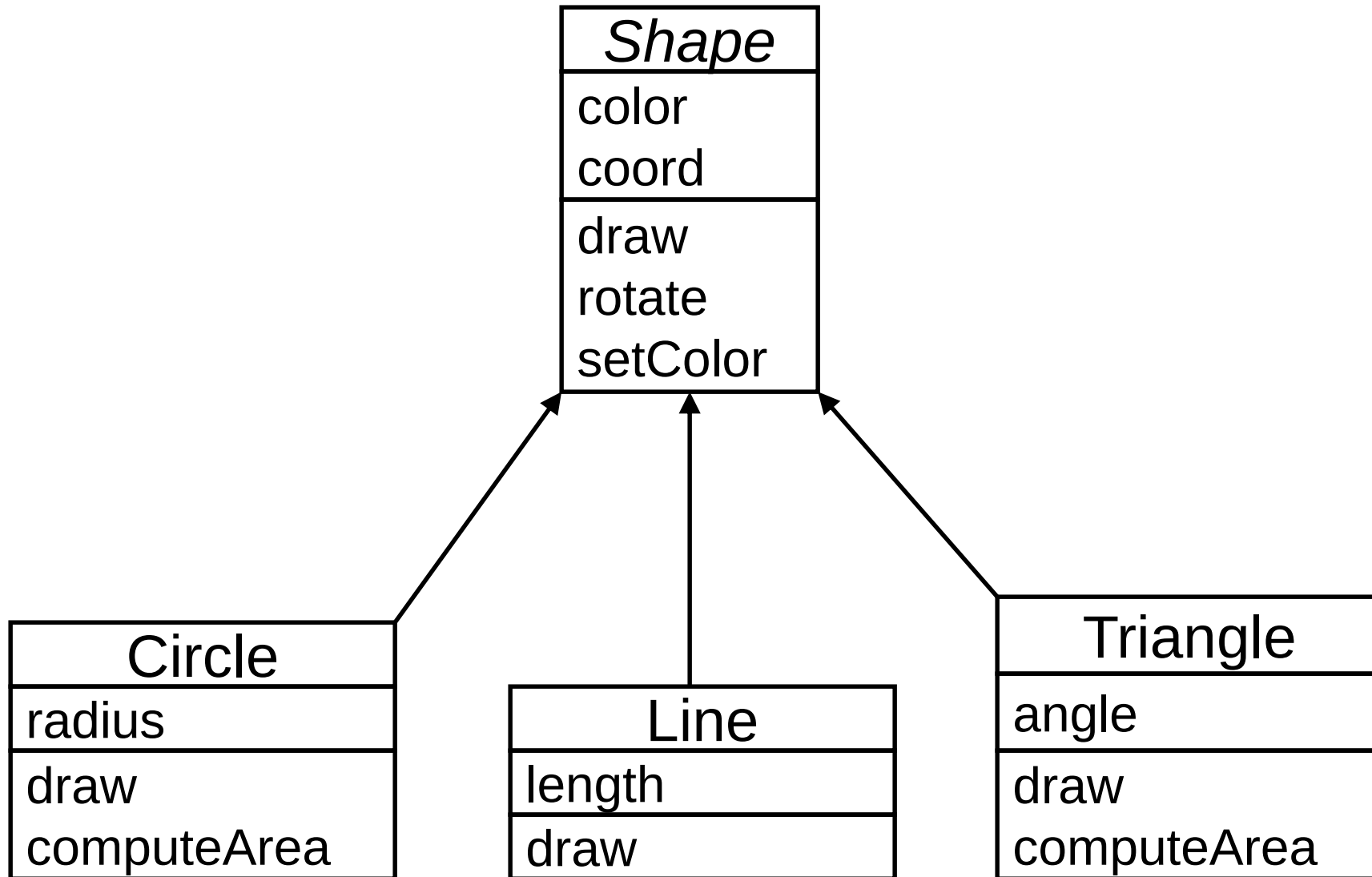
REUSE WITH INHERITANCE

Main purpose of inheritance is reuse

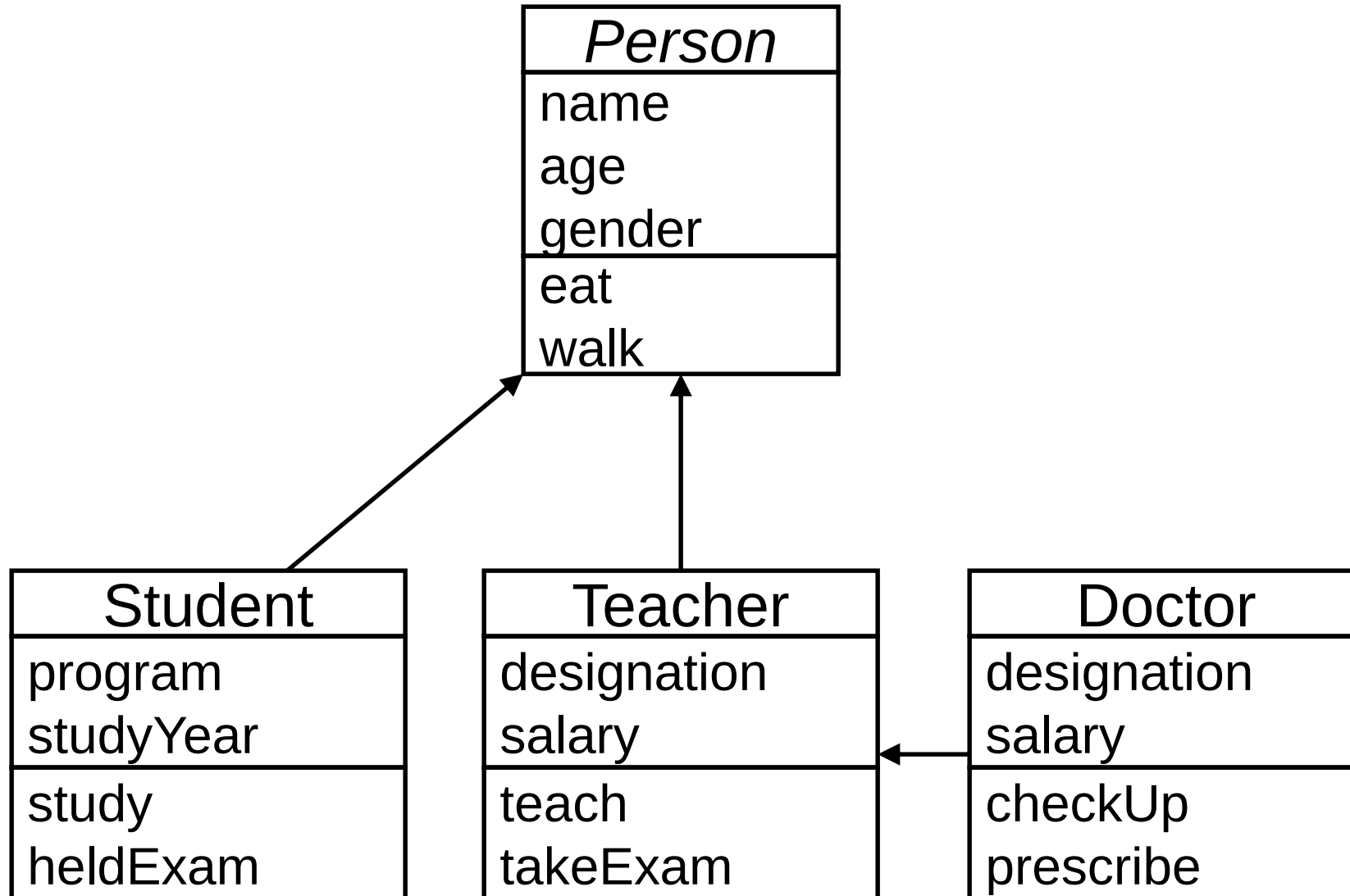
We can easily add new classes by inheriting from existing classes

- Select an existing class closer to the desired functionality
- Create a new class and inherit it from the selected class
- Add to and/or modify the inherited functionality

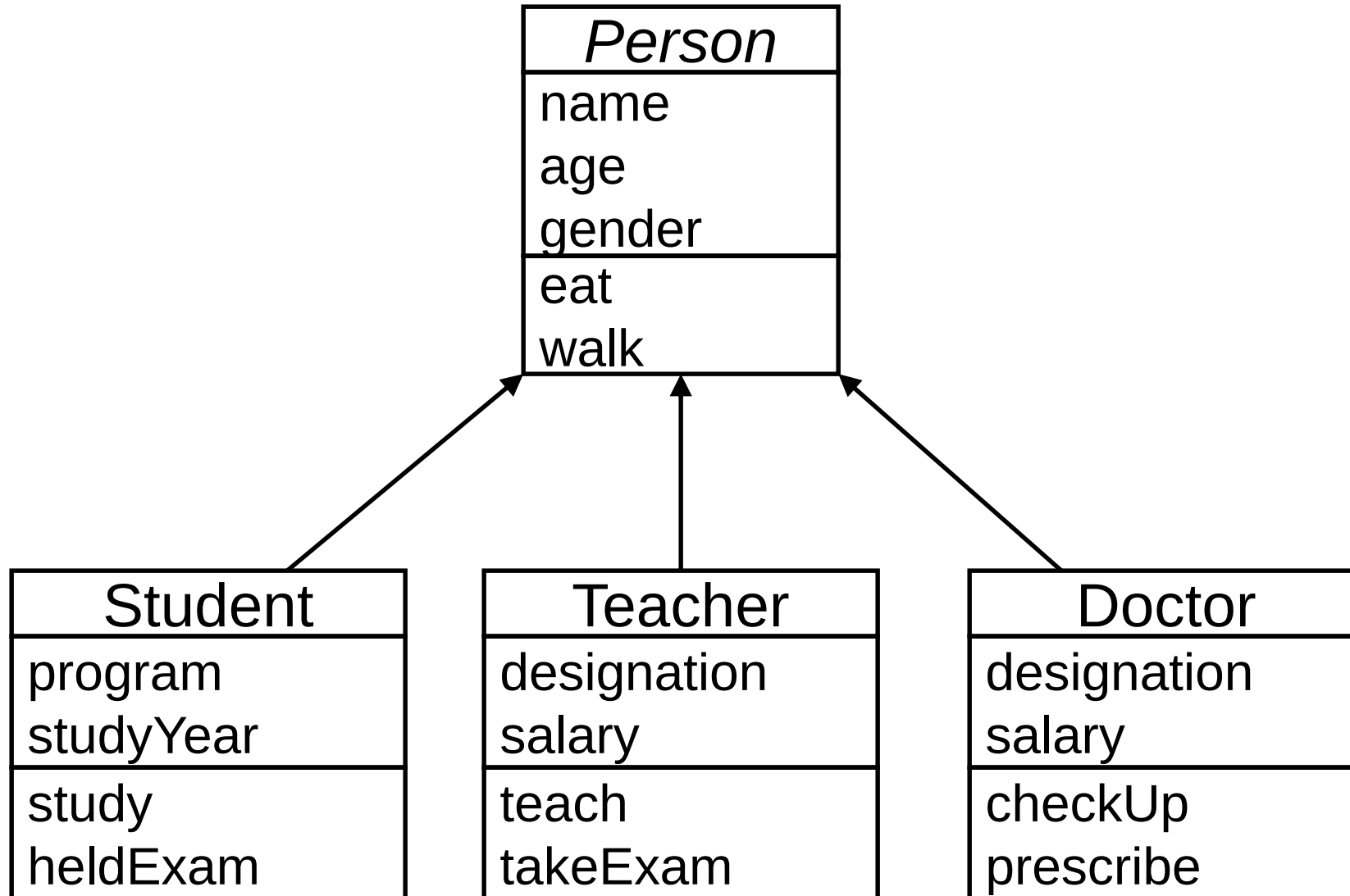
EXAMPLE REUSE



EXAMPLE REUSE

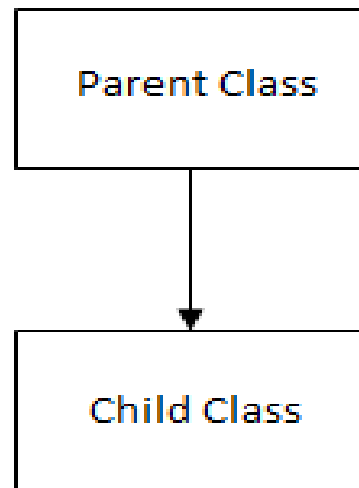


EXAMPLE REUSE

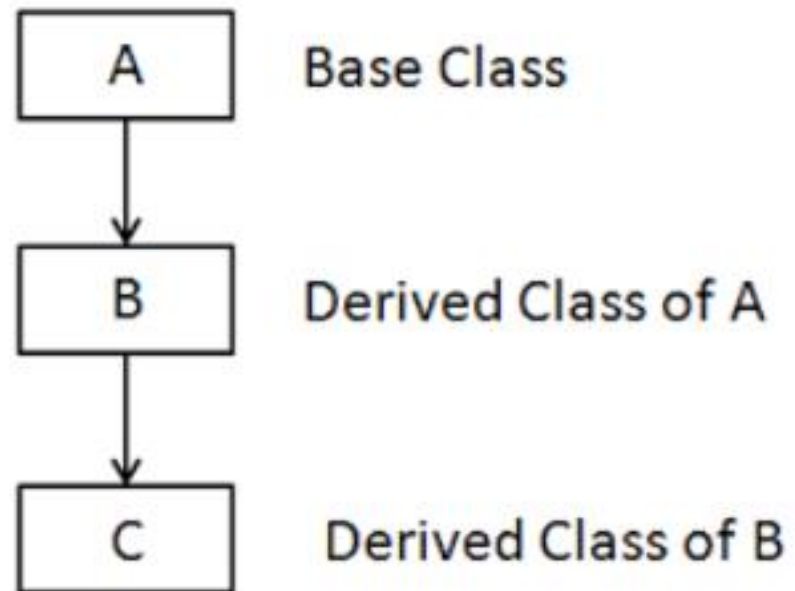


TYPES OF INHERITANCE

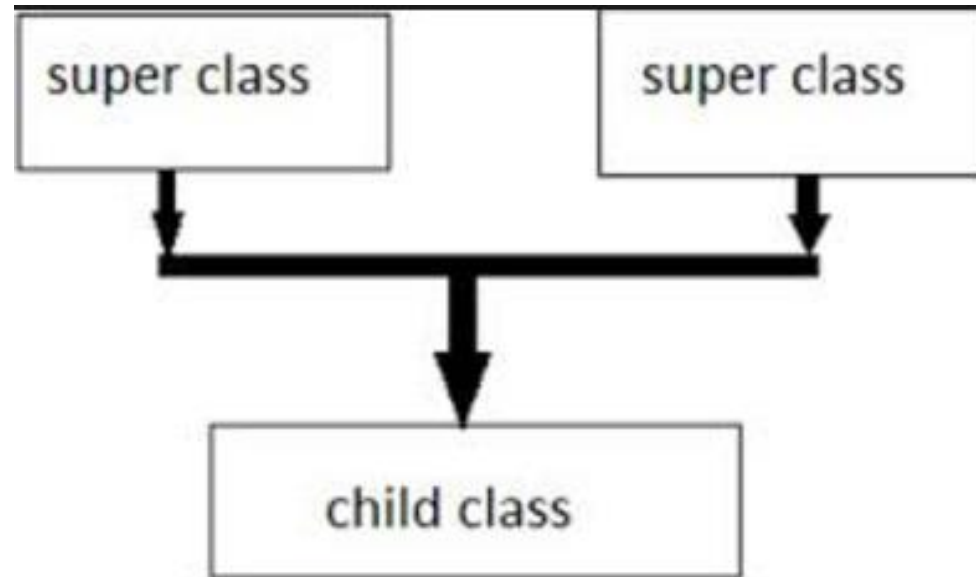
1 – Single Inheritance



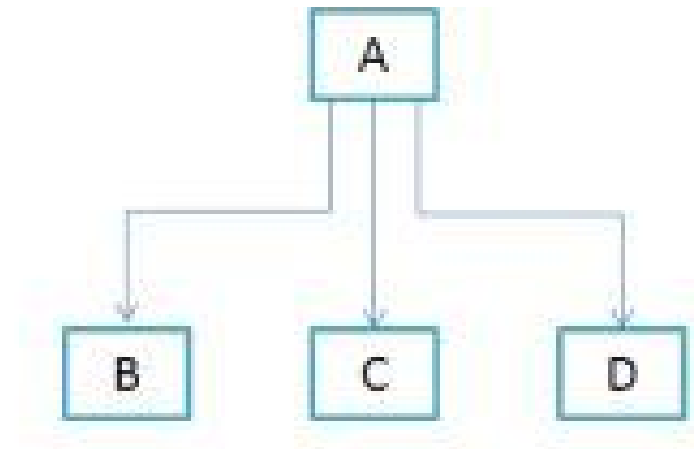
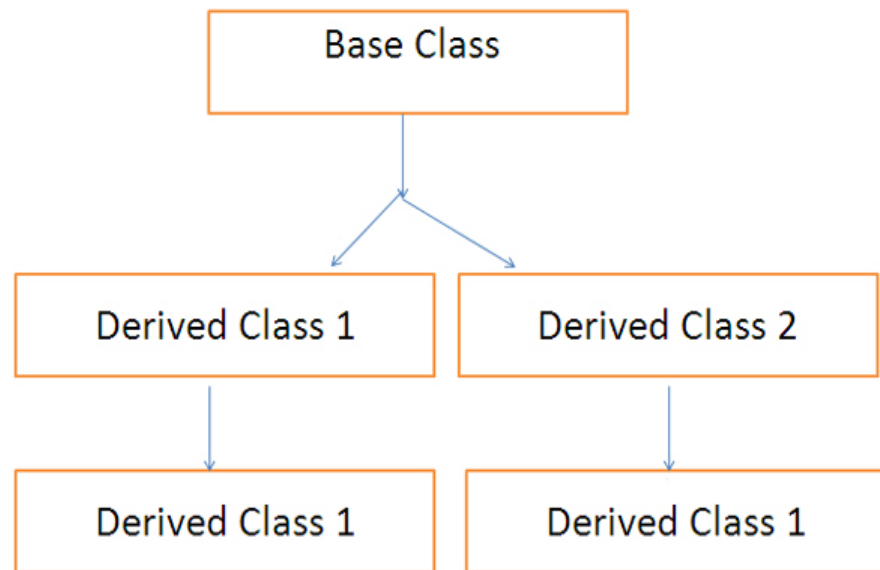
2 – Multi Level



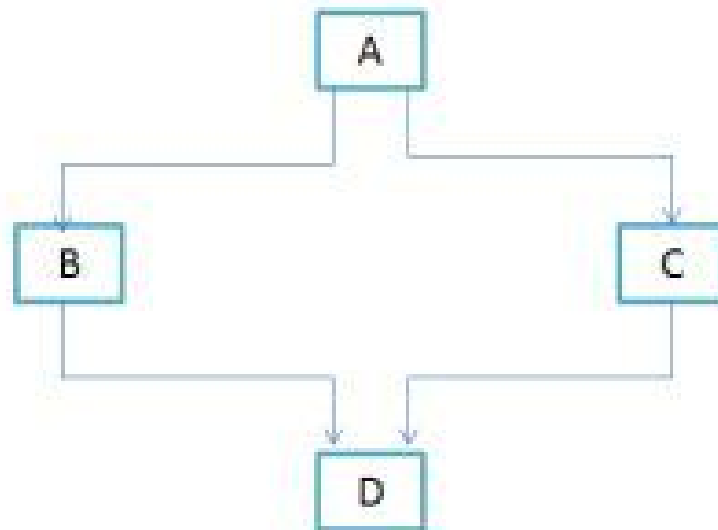
3 - Multiple



4 – Hierarchical



5 - Hybrid



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DEFINING A DERIVED CLASS WITH VISIBILITY MODE

Visibility Mode

1- Public

2- Private

3- Protected

Base Class Visibility	Derived class visibility		
	Public derivation	Private derivation	Protected derivation
Private	Not inherited	Not inherited	Not inherited
Protected	Protected	Private	Protected
Public	Public	Private	Protected